

NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:

- 1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W1_SEMI_NANO_RevA.STEP) GOVERNS THE NOMINAL GEOMETRY FOR ALL UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/- 0.03 mm, ANGULAR +/- 0.1 DEG (PER ISO 2768-f).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS ARE IN MILLIMETERS.

2. MATERIAL & MANUFACTURING PROTOCOL:

- 2.1 PRIMARY MATERIAL: TITANIUM ALLOY Ti-6Al-4V ELI (GRADE 23) PER ASTM F3001 / AMS 4930.
- MINIMUM YIELD STRENGTH $R_{p0.2} \geq 825$ MPa, MODULUS OF ELASTICITY $E = 114$ GPa.
- CALCULATED FINISHED MASS: 23.4 g.
2.2 FABRICATION ROUTE: PRECISION 5-AXIS CNC MACHINING / WIRE-EDM FROM BILLET STOCK (OR PBF-LB BULK NEAR-NET PREFORM FOLLOWED BY WEDM PROFILE FINISHING).
2.3 STRESS-RELIEF VACUUM ANNEALING PRIOR TO WIRE-EDM SEPARATION:
680°C +/- 10°C FOR 2 HOURS UNDER HIGH VACUUM (< 1.0E-4 mbar), CONTROLLED FURNACE COOLING TO INHIBIT INTERNAL STRESS DRIFT AND BLADE DISTORTION.

3. COMPLIANT JOINT & DfM MACHINING ATTRIBUTES:

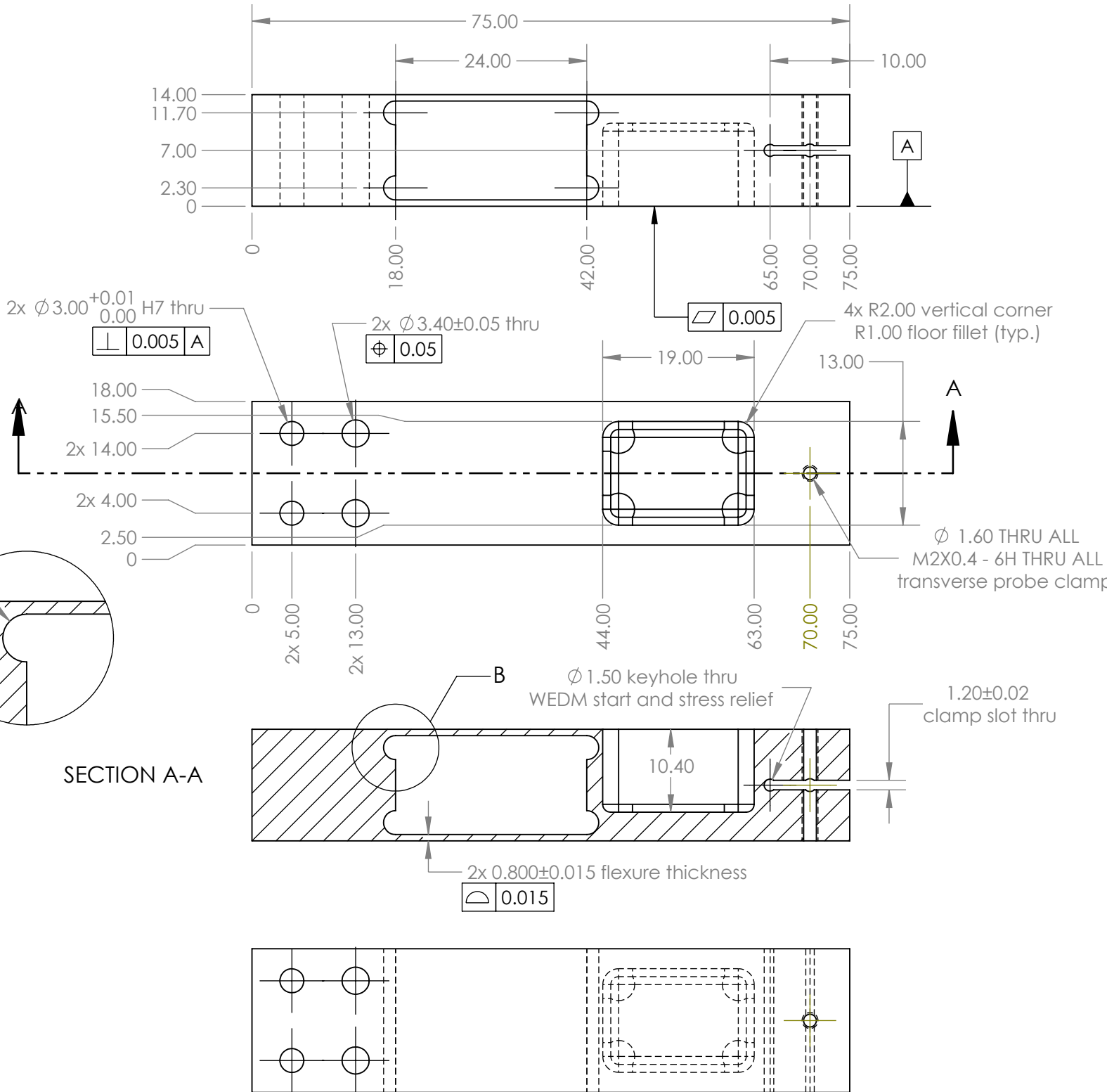
- 3.1 PARALLEL FLEXURE BLADE THICKNESS (0.80 +/- 0.015 mm):
UNIFORM DUAL-BLADE CONTINUOUS SECTION OVER 24.00 mm SPAN.
SURFACE PROFILE TOLERANCE ≤ 0.008 mm TO PREVENT BENDING HYSTERESIS.
3.2 FLEXURE ROOT FILLET RADII:
4X R 1.50 +/- 0.05 mm INTERNAL STRESS-RELIEF ARCS. NO SHARP TOOL NOTCHES OR STEP TRANSITIONS PERMITTED AT HINGE ROOTS.
3.3 LIGHTWEIGHTING POCKET (DfM CUTTING):
4X R 2.00 mm INTERNAL VERTICAL CORNERS TO SUIT DIA 3.0-3.5 mm END MILLS.
R 1.00 mm TRANSITION RADIUS AT CAVITY BOTTOM (STANDARD BULL-NOSE PROFILE).
3.4 PROBE TOOLING & CLAMPING INTERFACE:
- PROBE CLAMPING SLOT (1.20 +/- 0.02 mm GAP): CUT VIA WIRE-EDM (WEDM).
- SLOT RELIEF KEYHOLE (DIA 1.50 mm THRU): SERVES AS WEDM START HOLE AND STRESS CONCENTRATION ARRESTER.
- VERTICAL PROBE WIRE GUIDE (DIA 1.50 +/- 0.02 mm THRU).
- TRANSVERSE CLAMP SCREW PRE-DRILL (DIA 1.60 mm FOR M2-6H THREAD, 10.0 mm DEPTH).

4. KINEMATIC INTEGRITY & PERFORMANCE THRESHOLDS:

- 4.1 DATUM A: REAR/BOTTOM MOUNTING BASE FACE, FLATNESS ≤ 0.005 mm.
4.2 DATUM B: 2X DIA 3.000 +0.010/-0.000 mm (H7) DOWEL PINS, PERPENDICULARITY ≤ 0.005 mm TO [A].
4.3 Z-AXIS COMPLIANT DEFLECTION: 0.80 mm NOMINAL TRAVEL AT TIP UNDER 0.15 N CONTACT THRUST.
4.4 PARASITIC LATERAL SCRUBBING DRIFT (X/Y AXES): SHALL NOT EXCEED 0.05 MICRONS AT FULL STROKE.
4.5 DYNAMIC RESPONSE: FIRST NATURAL RESONANCE $f_1 \geq 4.2$ kHz; 40G SETTLING TIME ≤ 1.85 ms.

5. SURFACE PASSIVATION & CLEANROOM COMPLIANCE:

- 5.1 PASSIVATION: ACID DEGREASING & CHEMICAL PASSIVATION PER ASTM F86 (NITRIC ACID).
5.2 CRITICAL SURFACE ROUGHNESS:
- MOUNTING BASE [DATUM A]: $R_a \leq 0.4$ MICRONS.
- FLEXURE BLADES (INNER & OUTER): $R_a \leq 0.2$ MICRONS (ELECTROPOLISHED).
- TOOL CLAMP SLOT & WIRE GUIDE: $R_a \leq 0.4$ MICRONS.
5.3 CLEANLINESS STANDARD: ISO 14644-1 CLASS 3 COMPLIANT. ZERO METALLIC BURRS OR PARTICULATE SHEDDING UNDER 100X OPTICAL INSPECTION.
5.4 FINAL PACKAGING: PRECISION ULTRASONIC CLEANING IN DEIONIZED WATER (18 MOhm*cm), DRY-NITROGEN BLOWN, AND HERMETICALLY SEALED IN DOUBLE CLASS-100 CLEANROOM BAGS.



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: ISO 2768-f (Fine) LINEAR: ANGULAR:				FINISH: Ra 0.4 Flange & Base Ra 0.2 Flexures				DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION: A	
										CLEANROOM RATING: ISO 14644-1 Class 3			
	NAME		SIGNATURE		DATE					TITLE: MEMS NANO-PROBER COMPLIANT CANTILEVER ARM			
DRAWN										DWG NO. W1_SEMI_NANO_RevA			
CHK'D													
APPV'D													
MFG													
Q.A													
							MATERIAL:			A3			
							Titanium Ti-6Al-4V ELI (Grade 23) per ASTM F3001						
							WEIGHT: 23.4 g (Net Monolithic Solid)			SCALE: NTS		SHEET 1 OF 1	